

★ WEEK 3 · PROBLEM SET

Finance Applications: Three Hands-On Problems

Triple-Barrier · The Knobs (pt, sl) · Meta-Labeling & F1

Each problem: Statement → Principles & Lab Guide → Hand Calculation → Claude Code Prompt

Small numbers you can compute with a pencil — then a metaprompt that lets Claude Code scale the same idea to real market data. Hand first, machine second.

Every Problem Follows the Same Four Steps

Work top to bottom — do not skip the hand calculation, it is where the understanding lives

1 · Statement

the question

A concrete finance scenario with a tiny dataset you can see at a glance.

one slide

2 · Principles

the tools

The formulas involved and a short lab guide: what to compute, in what order.

one slide

3 · Hand Calc

pencil first

Work every number by hand in a table.
Small n , clean arithmetic, no calculator fear.

1–2 slides

4 · Claude Code

scale it up

A metaprompt: paste it into Claude Code and it builds the full lab on real data.

one slide

Hand first, machine second — you check the machine, never the other way around.

Where the Data Comes From

Public sources first — the prompts fetch free data; CSV files are offline fallbacks only

Problem	Primary source (public · no signup)	Provided fallback file
P1 · Triple-Barrier	yfinance — real daily prices (2y, AAPL)	fml_w3_prices_offline.csv
P2 · The knobs	yfinance — same series, three (pt, sl) settings	fml_w3_prices_offline.csv
P3 · Meta-labeling	yfinance + in-script momentum scout (fixed seed)	fml_w3_prices_offline.csv
Hand calculations	Printed on the slides (3 paths · 1 path × 3 knobs · 8 events)	fml_w3_p1 / p2 / p3 hand-calc CSVs

Course rule: if data is free online (yfinance, FRED, OpenML), the prompt fetches it — files are for offline classrooms.

P1

PROBLEM 1 • TRIPLE-BARRIER

Run the Race for Three Trades

Buy at 100, finish lines at 104 / 96, six days — first touch wins

Three Trades, One Set of Finish Lines

Profit-take +4%, stop-loss −4%, time limit 6 days. Label each path: +1, −1, or 0

Trade	Entry	D1	D2	D3	D4	D5	D6
A	100	101.0	102.0	103.0	104.5	—	—
B	100	99.0	100.0	98.0	96.5	95.8	—
C	100	101.0	100.0	102.0	101.0	103.0	102.0

The finish lines

- Profit-take = $100 \times 1.04 = 104$.
- Stop-loss = $100 \times 0.96 = 96$. Time limit = day 6.

Your deliverables

- First-touch day and label per trade.
- One sentence: why do B's later prices not matter?

The Tools: Two Lines, One Clock, One Rule

First touch wins — the rest of the path is never consulted

$$PT = 100 \times 1.04 = 104 \quad SL = 100 \times 0.96 = 96 \quad T = 6 \text{ days}$$

$$y = \begin{cases} +1 & \text{PT touched first} \\ -1 & \text{SL touched first} \\ 0 & \text{time barrier first} \end{cases}$$

Reading the rule

- Walk the path day by day, left to right.
- Compare each price to 104 and 96 — stop at the first touch.

Lab guide — per trade

- One row per day: price, vs profit-take, vs stop-loss, verdict. Touch → write the label and STOP the race. No touch by day 6 → label 0.

Walking the Three Races

Stop at the first touch — days after the touch are struck through mentally

Trade	Day-by-day vs lines (PT 104 / SL 96)	First touch	Label
A	101 · 102 · 103 all inside → D4: $104.5 \geq 104$	profit-take, day 4	+1
B	99 · 100 · 98 inside → D4: $96.5 \text{ still } > 96$ → D5: $95.8 \leq 96$	stop-loss, day 5	-1
C	101 · 100 · 102 · 101 · 103 · 102 — never touches either line	time limit, day 6	0

$$y_A = +1 \text{ (day 4)} \quad y_B = -1 \text{ (day 5)} \quad y_C = 0 \text{ (day 6)}$$

One of Each — and a Lesson in B

The label is decided at the touch; everything after is another trade's story

+1

Trade A — profit-take touched day 4; days 5–6 never checked

−1

Trade B — stop-loss touched day 5; any later rebound is irrelevant

0

Trade C — six days of wandering; the clock wins the race

Why B's future doesn't matter

- A real stop-loss order closes the trade at 96.
- The label records what the trader lived — not the counterfactual.

Contrast: the lazy label

- 'Up after 6 days?' would label B by its day-6 price alone.
- Path-blindness is exactly the trap we came here to fix.

Scale It: From Three Races to Two Years

Paste the metaprompt — Claude Code labels every day of a real price series

[INPUT] Prompt · Claude Code

```
Build a single Colab-ready Python script that:  
1) downloads 2 years of daily AAPL prices with yfinance  
   (auto_adjust=True);  
   (offline classroom: load fml_w3_prices_offline.csv)  
2) labels every day with a Triple-Barrier race:  
   pt = +2%, sl = -2%, time limit = 10 days,  
   first touch wins (+1 / -1 / 0);  
3) prints the count and share of each label and plots  
   the price with days colored by their label;  
4) also reports the average number of days until the  
   first touch, per label class;  
5) one-line verdict: is the class mix usable for ML,  
   or do the lines need retuning?
```

[OUTPUT] What to expect

```
roughly 40% +1, 35% -1,  
25% 0 (varies by period)  
+1/-1 touches arrive in a  
few days; 0s take all 10  
colored stretches follow  
trends and selloffs
```

```
Your three hand races,  
repeated ~500 times.
```

P2

PROBLEM 2 • THE KNOBS

Same Path, Three Different Answers

One price path through three (pt, sl) settings — the label is a choice

One Path, Three Rulebooks

Run the SAME six days through (+2,-2), (+3,-1) and (+1,-3). Time limit: 6 days

Day	Entry	D1	D2	D3	D4	D5	D6
Price	100.0	101.5	99.2	102.2	98.8	103.2	99.5
Return from entry	0.0%	+1.5%	-0.8%	+2.2%	-1.2%	+3.2%	-0.5%

The three rulebooks

- Neutral: pt +2%, sl -2%.
- Trend style: pt +3%, sl -1%.
- Quick-profit style: pt +1%, sl -3%.

Your deliverables

- First touch and label under each rulebook.
- One sentence per rulebook: what philosophy did it encode?

The Tool: The Label Is a Function of Your Knobs

Nothing about the market changes — only the question you ask it

$$y = y(\text{path}; \text{pt}, \text{sl}, T) \quad \text{same path, different knobs} \Rightarrow \text{different label}$$

Why this matters

- Two researchers, same data, different knobs $\rightarrow$ different 'truths'.
- The knob ratio pt:sl encodes a trading philosophy (Lecture slide 15).
- Always report the knobs next to the label counts.

Lab guide — per rulebook

- Walk D1 $\rightarrow$ D6 comparing the return-from-entry to pt and sl.
- First crossing decides; no crossing by D6 $\rightarrow$ label 0.
- Three quick passes over the same row of numbers.

Three Passes Over One Path

Watch the same +1.5% day be 'nothing' in one rulebook and a WIN in another

Rulebook (pt, sl)	Walk (return from entry)	First touch	Label
Neutral (+2, -2)	+1.5 no · -0.8 no · +2.2 $\geq$ +2 ✓	D3 — profit-take	+1
Trend (+3, -1)	+1.5 no · -0.8 no · +2.2 no · -1.2 $\leq$ -1 ✓	D4 — stop-loss	-1
Quick-profit (+1, -3)	+1.5 $\geq$ +1 ✓ on the very first day	D1 — profit-take	+1

$(+2, -2): y = +1$ (day 3) $(+3, -1): y = -1$ (day 4) $(+1, -3): y = +1$ (day 1)

Three Rulebooks, Three Verdicts

+1 · -1 · +1 from identical prices — the knobs ARE the question

+1 (D3)

Neutral: a clean +2.2% follow-through

-1 (D4)

Trend style: the tight -1% stop fired before +3% arrived

+1 (D1)

Quick-profit: +1.5% on day one banked instantly

0 changes

The market did nothing different in the three passes

The uncomfortable lesson

- 'Ground truth' in financial ML is partly designed, not found.
- A model trained on trend-style labels learns a trend-style game.

The discipline

- Choose knobs to match the strategy you would actually run.
- Then keep them fixed — retuning knobs to improve results is peeking.

Scale It: Sweep the Knob Grid

Paste the metaprompt — Claude Code maps how the label mix responds to every setting

[INPUT] Prompt · Claude Code

Build a single Colab-ready Python script that:

- 1) downloads 2 years of daily AAPL prices with yfinance;
(offline classroom: load fml_w3_prices_offline.csv)
- 2) implements the Triple-Barrier labeler from class
(pt, sl, 10-day limit, first touch wins);
- 3) sweeps a grid: pt in {1%, 2%, 3%} x sl in
{-1%, -2%, -3%} and computes the +1 / -1 / 0 shares
for all nine combinations;
- 4) prints the nine rows as a table and draws a heatmap
of the +1 share over the grid;
- 5) three-sentence verdict: which corner of the grid is
'let winners run', which is 'take quick profits',
and why a model must never tune these knobs by
choosing whatever maximizes backtest profit.

[OUTPUT] What to expect

(+3,-1): -1 share jumps -
the tight stop dominates
(+1,-3): +1 share jumps -
quick targets get hit
(+2,-2): the balanced row

Your one-path table, run
over ~500 paths and nine
rulebooks at once.

P3

PROBLEM 3 · META-LABELING

The Scout, the Manager, and F1

Eight signals, one filter — precision 50→80%, F1 67→89%, by hand

A Momentum Scout and Its Track Record

Eight signals: the side it chose, its conviction $|\text{mom}|$, and what the barrier race said

Event	Scout side	Momentum	Barrier outcome
E1	LONG	+3.0%	+1
E2	LONG	+1.0%	-1
E3	SHORT	-2.5%	-1
E4	LONG	+2.0%	-1
E5	SHORT	-1.0%	+1
E6	LONG	+0.5%	-1
E7	SHORT	-3.0%	-1
E8	LONG	+2.5%	+1

Your deliverables

- Meta-label per event: 1 if the side matched the outcome, else 0.
- Precision, recall and F1 taking ALL eight bets.
- The same three numbers taking only $|\text{mom}| \geq 2\%$ bets.
- One sentence: what did the manager actually remove?

A LONG is right when the outcome is +1; a SHORT is right when it is -1.

The Tools: One Indicator, Three Ratios

The meta-label is 'was the scout right?' — precision, recall and F1 score the manager

$$\text{meta-label}_i = \mathbf{1}\{\text{primary side}_i \text{ was right}\} \quad \text{Precision} = \frac{\text{correct bets}}{\text{bets taken}}$$

$$F_1 = \frac{2PR}{P+R}$$

The hand-made 'manager'

- Real managers are logistic regressions on features.
- Today's stand-in: bet only if $|\text{mom}| \geq 2\%$ —

Lab guide — three steps

- Step 1: mark each event correct (1) or wrong (0). Step 2: precision, recall, F1 over all 8 bets. Step 3: drop $|\text{mom}| < 2\%$ rows and recompute all three — recall counts the correct bets you KEPT out of all 4 correct ones.

Scoring the Eight Bets

The manager keeps E1, E3, E4, E7, E8 — and fires the three weakest signals

Event	Side / outcome	Correct?	$ \text{mom} \geq 2\%$	Manager's verdict
E1	LONG / +1	1	yes (3.0)	kept — correct
E2	LONG / -1	0	no (1.0)	fired
E3	SHORT / -1	1	yes (2.5)	kept — correct
E4	LONG / -1	0	yes (2.0)	kept — wrong
E5	SHORT / +1	0	no (1.0)	fired
E6	LONG / -1	0	no (0.5)	fired
E7	SHORT / -1	1	yes (3.0)	kept — correct
E8	LONG / +1	1	yes (2.5)	kept — correct

All three fired bets were wrong; only one wrong bet (E4) slipped through the filter.

Precision 50→80%, Recall Held at 100%, F1 67→89%

The manager never touched direction — it just refused the weakest hands

$$\text{all bets: } \frac{4}{8} = 50\% \quad \longrightarrow \quad \text{filtered } (|\text{mom}| \geq 2\%) : \frac{4}{5} = 80\%$$

$$\text{all bets: } F_1 = \frac{2(0.50)(1.00)}{1.50} = 66.7\% \quad \text{filtered: } F_1 = \frac{2(0.80)(1.00)}{1.80} = 88.9\%$$

Reading the three ratios

- Precision: $4/8 \rightarrow 4/5$ — fewer bets, better bets.
- Recall: $4/4$ kept — the filter fired no winners.
- F1: $66.7\% \rightarrow 88.9\%$ — both halves honest, so the blend jumps.

From rule to model

- Replace ' $|\text{mom}| \geq 2\%$ ' with logistic regression on $[|\text{mom}|, \text{vol}]$.
- Its output $P(\text{scout right})$ sizes the bet — the micro-case slide.
- That upgrade is exactly what the Claude prompt builds next.

Scale It: A Real Manager on Real Signals

Paste the metaprompt — Claude Code fits the logistic manager and scores the gap

[INPUT] Prompt · Claude Code

```
Build a single Colab-ready Python script that:  
1) downloads 2 years of daily AAPL prices with yfinance  
   and labels every day with the Triple-Barrier race  
   (pt +2%, sl -2%, 10-day limit);  
   (offline classroom: load fml_w3_prices_offline.csv)  
2) builds a SCOUT: side = sign of the 20-day momentum;  
   drop 0-label days; meta-label = 1 if side matched  
   the barrier label;  
3) fits sklearn LogisticRegression on features  
   [abs(momentum), 20-day volatility] - 70/30  
   TIME-ORDERED split, no shuffling;  
4) on the test set prints precision, recall and F1 of:  
   all scout bets vs bets the manager approves (P>0.5);  
5) three-sentence verdict in scout-and-manager language,  
   citing which ratio moved most and why.
```

[OUTPUT] What to expect

```
scout alone: precision  
~ 50 - 60%, F1 modest  
manager-filtered: precision  
clearly higher, recall  
dips a little, F1 rises  
  
Your 8-event table with a  
learned cutoff instead of  
the hand rule |mom| >= 2%.
```

Problem Set — Three Things Your Hands Now Know

You computed every number yourself; the code only repeats what you already understand

1

Triple-Barrier: first touch wins the race

A(+1, day 4), B(-1, day 5), C(0) — and B's rebound never mattered, just like a real stop.

2

The knobs ARE the question

One path gave +1 / -1 / +1 under three rulebooks — the label encodes a trading philosophy.

3

The manager: precision 50→80%, F1 67→89%

It fired three weak, wrong bets and kept every winner — direction untouched, size disciplined.

Hand calculation first — then let Claude Code scale the same idea to real data.